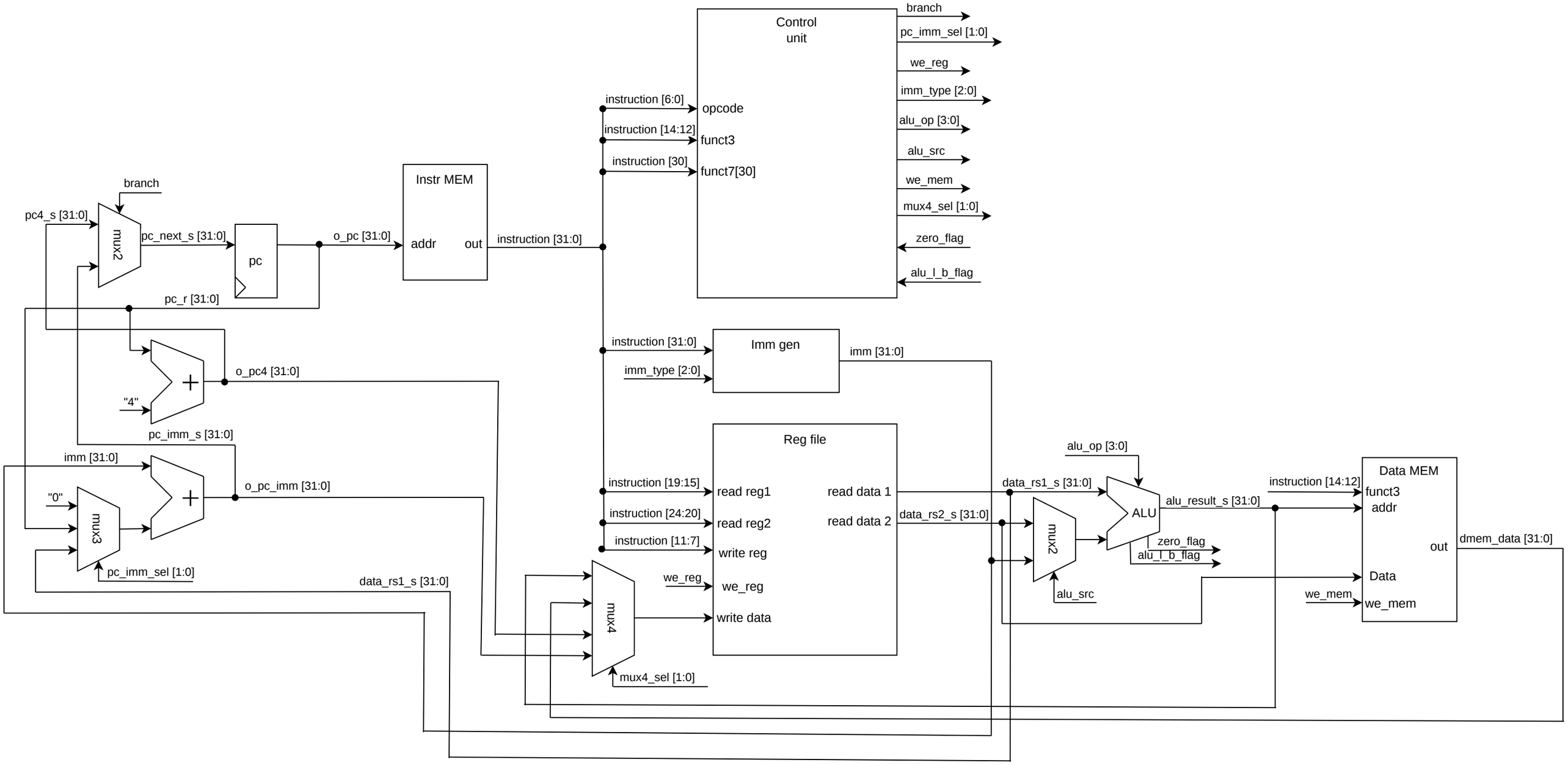
Control
unit
branch
pc_imm_sel [1:0]
we_reg
imm_type [2:0]
alu_op [3:0]
alu_src
we_mem
mux4_sel [1:0]
zero_flag
alu_l_b_flag
instruction [6:0]
opcode
instruction [14:12]
funct3
instruction [30]
funct7[30]
Instr MEM
addr
out
instruction [31:0]
branch
pc4_s [31:0]
mux2
pc_next_s [31:0]
pc
o_pc [31:0]
pc_r [31:0]
"4"
o_pc4 [31:0]
Imm gen
instruction [31:0]
imm_type [2:0]
imm [31:0]
pc_imm_s [31:0]
imm [31:0]
"0"
mux3
o_pc_imm [31:0]
pc_imm_sel [1:0]
Reg file
instruction [19:15]
read reg1
read data 1
instruction [24:20]
read reg2
read data 2
instruction [11:7]
write reg
we_reg
we_reg
write data
data_rs1_s [31:0]
mux4
mux4_sel [1:0]
alu_op [3:0]
data_rs1_s [31:0]
data_rs2_s [31:0]
mux2
ALU
alu_result_s [31:0]
zero_flag
alu_l_b_flag
alu_src
instruction [14:12]
Data MEM
funct3
addr
out
Data
we_mem
we_mem
dmem_data [31:0]